Optimization of Glass Core Substrate Design for Energy-Efficient High-Density AI Server Packages

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Abstract

A heterogeneous AI server package integrating XPU–HBM chiplets and co-packaged optics (CPO) on a glass core substrate is presented. This work addresses system-level mechanical, electrical, and power co-optimization for multi-module integration on glass, rather than substrate process comparison, by evaluating CTE-induced warpage, TGV-based power delivery, electromigration robustness of inorganic-coated RDL layers, and high-speed signal transmission between XPU and CPO modules. The results demonstrate improved signal integrity, power efficiency, and reliability for large-scale, energy-efficient AI server packages.

Author Keywords

Glass core substrate Warpage analysis Co-packaged optics (CPO) Through-glass via (TGV) Signal and power integrity

Introduction

The rapid growth of AI data servers demands higher computing performance under strict power consumption constraints, making co-packaged optics (CPO) an essential technology to reduce electrical I/O power dissipation.

In parallel, AI accelerators integrating XPUs and high-bandwidth memory (HBM) continue to increase in chiplet count and die size, driving package dimensions toward approximately 200 mm square by the 2030s.

Such large-area packages pose critical challenges in mechanical stability and warpage control, where conventional organic core substrates become insufficient, positioning glass core substrates as a key enabler due to their low CTE and high dimensional stability. In this work,

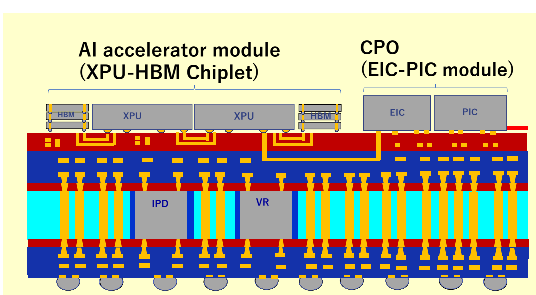

Figure 1. Energy Efficient PKG for AI data server

we propose a next-generation AI server package architecture based on a glass core substrate with advanced build-up interconnects, as illustrated in Figure 1.

The build-up layers incorporate a high-speed wiring structure for low-loss XPU–CPO transmission and a fine-pitch interconnect structure for XPU–HBM connectivity.

An inorganic protective coating is applied to the fine wiring layers to enhance electromigration robustness under high current density. The proposed architecture is evaluated in terms of warpage resistance, TGV-based power delivery capability, electromigration reliability, and high-speed signal transmission, demonstrating its suitability for large-scale, energy-efficient AI server packages.

Package size and warpage

According to Yole (2025), the substrate size of AI accelerator packages is expected to reach 120 × 120 mm² by 2028, highlighting the challenge of rapidly increasing form factors. This trend indicates further package size expansion when CPO modules are integrated within the same package.

Figure 2 shows reflow warpage as a function of package outer size evaluated using a simplified material–thickness model. The warpage model is derived from classical thermo-mechanical bending theory (Timoshenko bi-material formulation) combined with plate theory, yielding a first-order scaling relationship $W \propto (\Delta CTE/E) \cdot (L2/t)W$.[1] Results for organic cores (CTE = 16 and 5 ppm/K) and a glass core (E = 64 GPa, CTE = 3.3 ppm/K, t = 1.0 mm) are compared. JEITA allowable warpage limits are indicated for reference. The glass core substrate maintains warpage within JEITA limits for large package sizes.

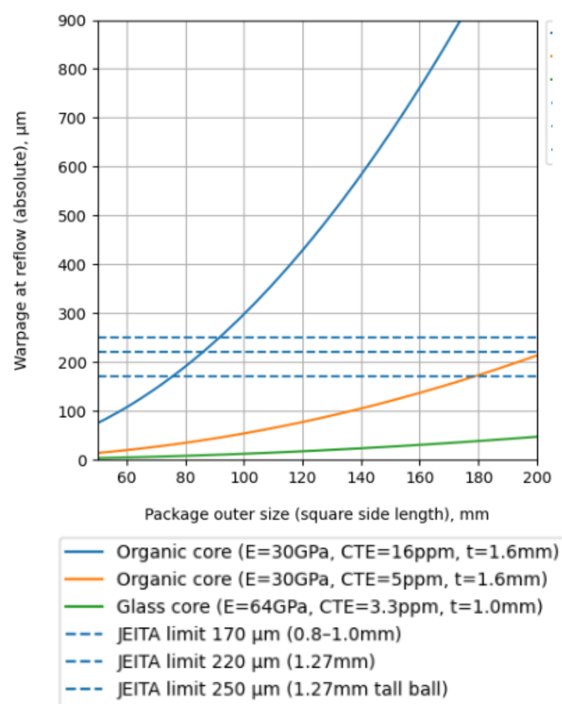

Figure 2. Reflow warpage as a function of package outer size

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Power delivery and TGV specification

Table 1. Required power density

		2022	2024	2026	2028	2030	2032	
Application		Switch、HPC、AI						
DC power consumption	TWh	415	520	680	830	945	1050	
Technology node		12nm	7-5nm	3-2nm		1.5nm		
Logic structure		finFET		GAA	CFET-SRAM			
Maximum chip power	W/mm^2		5	7	10	14	20	※1
Transistor supply voltage	V		1	0.7	0.65	0.6	0.6	※2
Current density standard	A/mm^2		5	10	15.4	23.4	33.4	

※1 HIR Roadmap
※2 IRDS

Table 1 summarizes the required power density for logic devices across technology generations, indicating a significant increase driven by device scaling and architectural evolution. This trend motivates high-capacity, low-loss power delivery at the package level.
Power delivery through metallized through-glass vias (TGVs) was analyzed using current-density simulations (Figure 3(a)) and maximum-current scaling with via cross-sectional area (Figure 3(b)). Glass substrates enable fine-pitch via fabrication, allowing multi-TGV configurations that provide high current capacity suitable for next-generation AI accelerators.

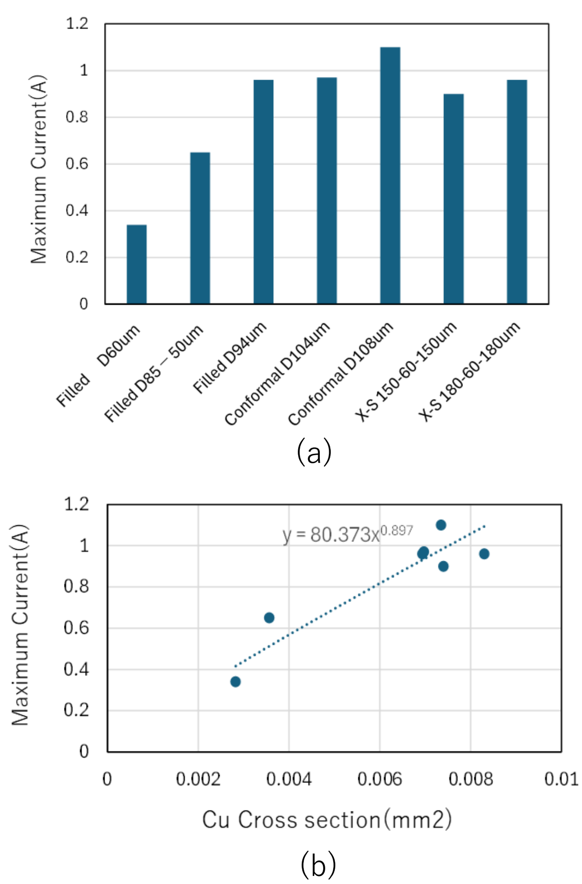

Figure 3. Maximum currents depend on Cross section area

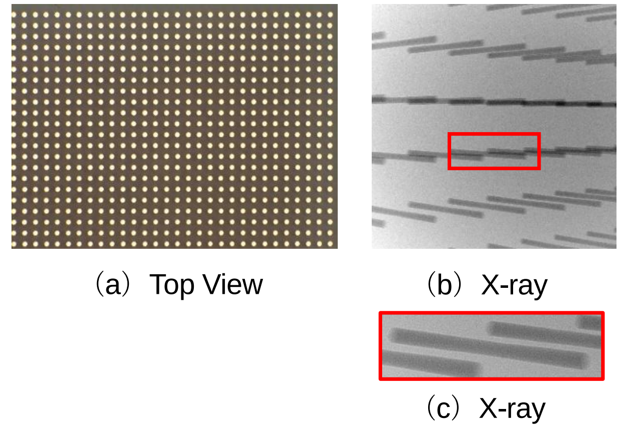

Figure 4. Fine pitch 100um TGV

A fine-pitch, metallized TGV implementation with 50 µm diameter 100µm pitch is shown in Figure 4(a)(b)(c).

Fine pitch RDL architecture

Figure 5 illustrates the fine-pitch RDL design employing an inorganic cover layer. [2] Inorganic-passivated Cu (IP-Cu) enhances reliability, and coplanar waveguide (CPW) transmission lines were fabricated to evaluate high-frequency characteristics. Measured results for 5-mm-long CPW lines are shown in Figure. 6. Insertion loss increases with finer wiring pitch due to increased conductive loss associated with reduced cross-sectional area.[3]

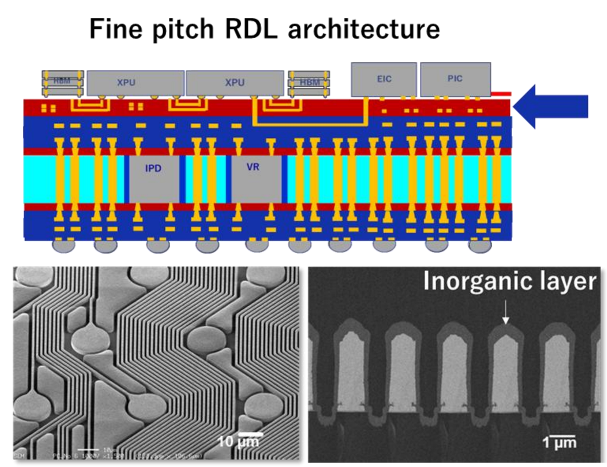

Figure 5. Fine pitch RDL architecture with inorganic layer.

Figure 7 shows and demonstration of eye diagram. [4] Figure 7 demonstrate after TCT test insist that Inorganic layer enhance high speed transmission properties.

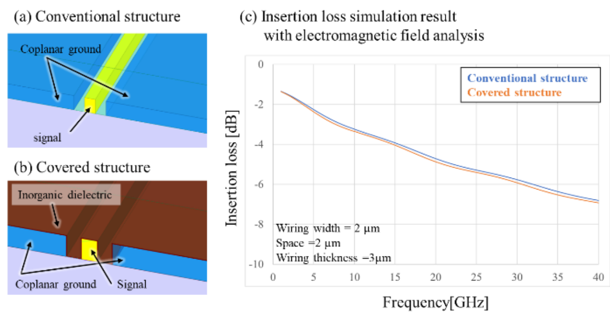

Figure 6. Measurement result of fine wiring consists with CPW structure.

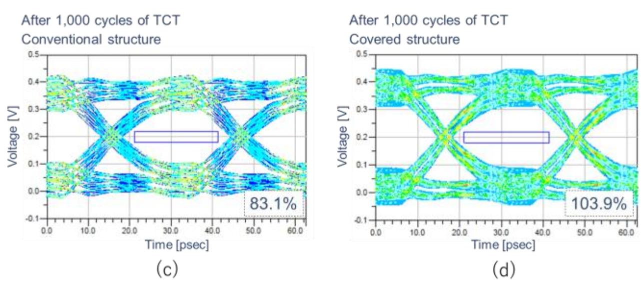

Figure 7. Eye diagram simulation result of CPW on RDL after 1000cycle of TCT.

Electromigration (EM) performance of fine-pitch RDLs is presented in Figure 8[5], demonstrating improved lifetime compared with conventional Cu RDLs. Activation energies extracted using Black's equation are summarized in Table 3.

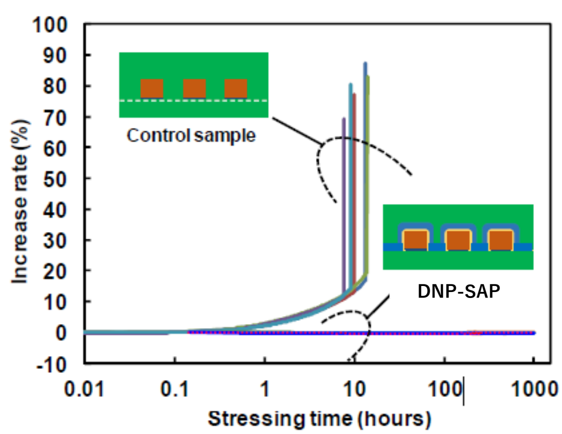

Figure 8. Increase rate of electrical resistance of 1-μm-wide Cu traces stressed with a current density of 2.0E6 A/cm2.

Table3 Activation Energy calculated by Blacks equation

Condition	Line Width (μm)	Current Density (A/cm²)	Joule Temp (° C)	Time to Failure – conventional(h)	Time to Failure – DNP SAP (h)	Activation Energy – conventional(eV)	Activation Energy – DNP SAP (eV)
A	1	2.00E+06	180	10	1000	0.913	1.093
B (Ref.)	2	2.00E+06	180	7	1000	0.913	1.107
C	2	2.50E+06	230	0.7	1000	0.913	1.228

High speed RDL architecture

High-speed RDLs are required to connect AI chiplet modules and CPO modules within energy-efficient packages. The proposed high-speed RDL architecture is shown in Figure 9. Low insertion loss was demonstrated for XPU–CPO interconnects at 70 and 100 GHz, as shown in Figures 10. Insertion loss depends on cross-section trace of edge show in Figure 11.

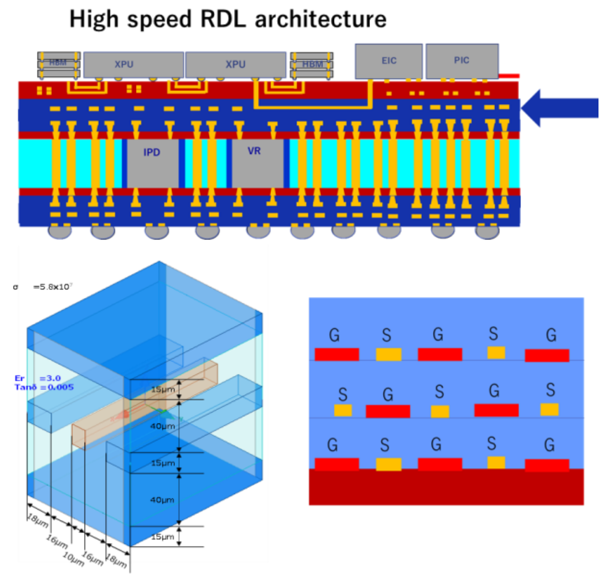

Figure 9. High speed RDL architecture

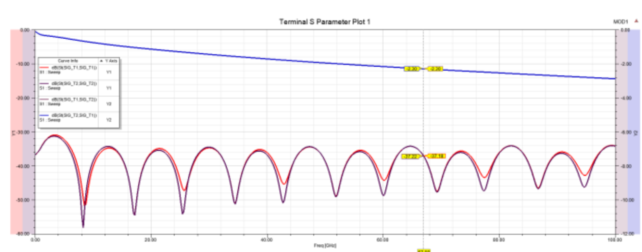

Figure 10. Transmission line characteristics between AI module and CPO module

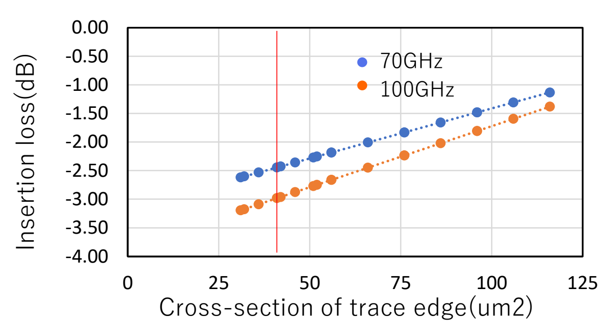

Figure 11. Insertion loss depends on cross-section trace of edge

Reliability result of Glass substrate

Figure 12 shows a build-up substrate with two RDL layers fabricated on a glass core (panel size: 300 × 400 mm). Cu-filled TGVs with a diameter of 100 µm were arranged at a 1 mm pitch, resulting in approximately 24,000 vias within a 160 × 160 mm² area. Reliability was evaluated using substrates with a resin buffer layer. After preconditioning, thermal shock testing (−55 °C to 150 °C), and HAST, SEWARE was not observed in samples with the buffer layer when three and six ABF layers were laminated. (Figure 13, Table4).

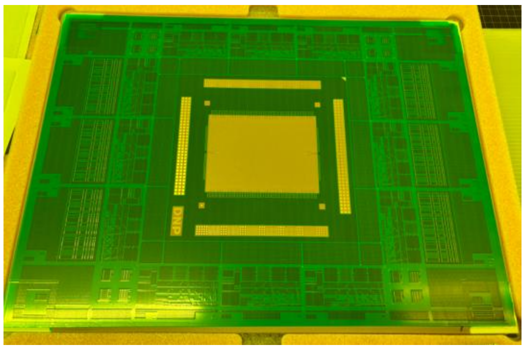

Figure 12. 160mmX160mm2 RDL on Glass substrate

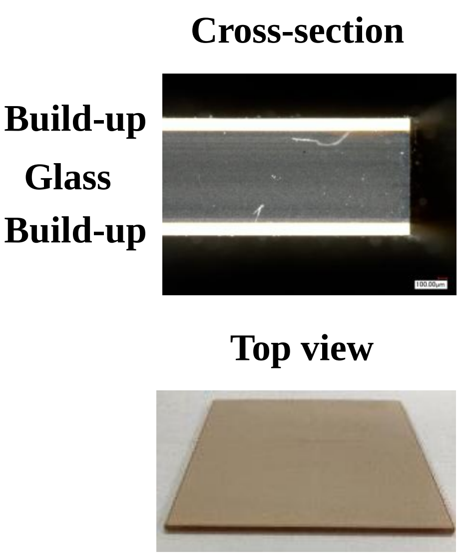

Figure 13. Reliability test result

Table 4 reliability test result of Glass core substrate

Sample Structure	ABF	Process applied for verification			
		Dicing	*Precondition*	*HAST*	*TST*
with resin buffer layer	±3 layers	Pass	Pass	Pass	Pass
	±6 layers	Pass	Pass	Pass	Pass

4.Conclusion

Key core substrate properties—including via pitch, insertion loss, power density, elastic modulus, and CTE—were identified as critical factors for large-scale AI server packages. Warpage simulations incorporating complex chiplet and glass core properties revealed a strong dependence on core modulus and CTE. RDLs with inorganic protective layers enhanced both reliability and high-speed signal transmission for XPU–HBM interconnects, while low insertion loss was demonstrated for XPU–CPO links at 70 and 100 GHz. In addition, a 160 × 160 mm² RDL structure on a glass core substrate with Cu-filled TGVs was successfully fabricated on a 300 × 400 mm² panel, and a resin buffer layer was shown to be essential for suppressing SEWARE and ensuring long-term reliability.

5.Reference

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